Application Note

Post Processing, Angle and INL Calculation

for Analog Output Signals

v0-01 • 2020-Jun-29

Content Guide

1 Introduction

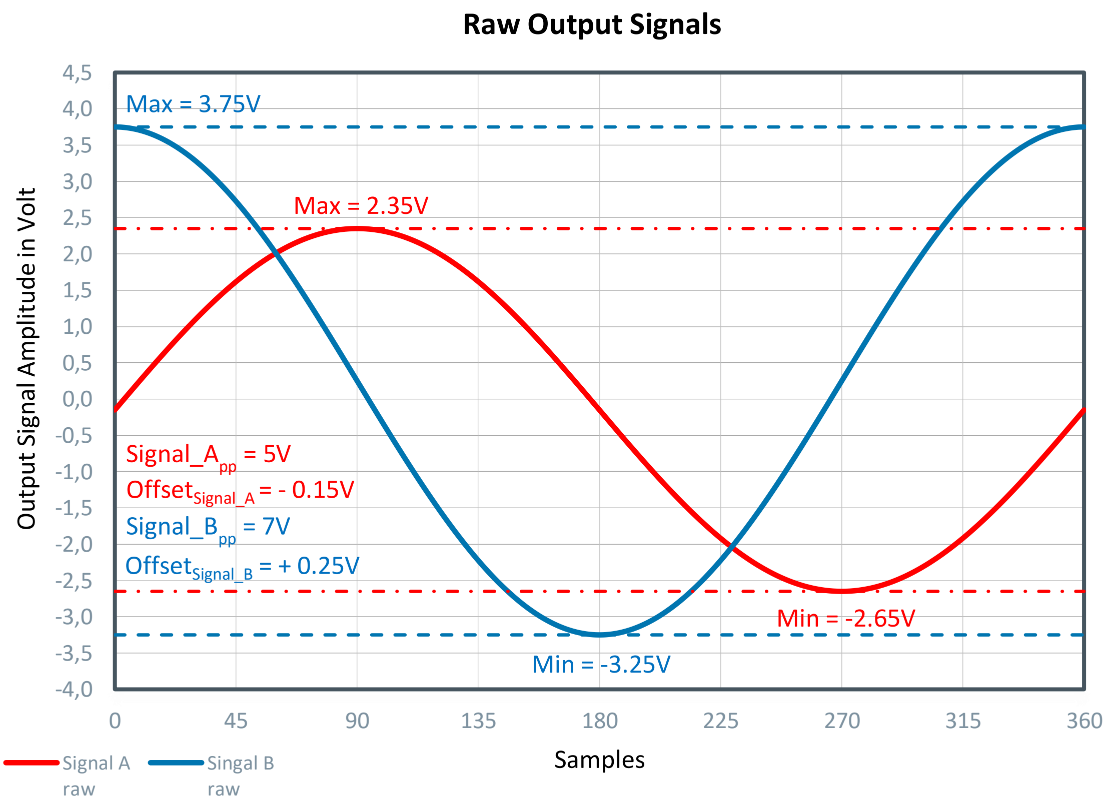
Depending on the application, it sometimes occurs that the analog output signals of a sensor IC have different amplitudes and are afflicted by an offset. The purpose of this document is, to give an overview of how to normalize gain and offset of analog output signals. Furthermore, it provides information, of how to calculate an angle and the corresponding INL out of two sinusoidal shaped signals.

2 Gain and Offset Correction

2.1 Raw Output Signals

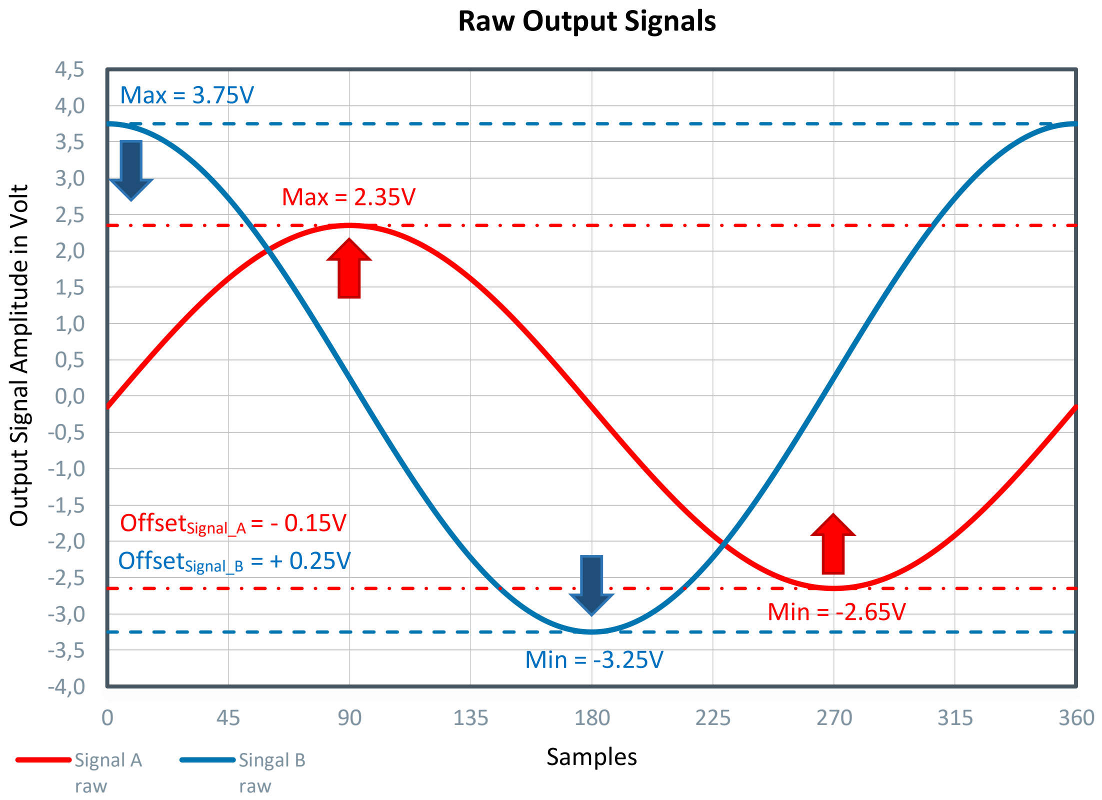
Figure 1 shows two exemplary analog output signals, Signal A and Signal B, which have different amplitudes and are afflicted by an offset.

Figure 1:
Raw Output Signals



2.2 Offset Correction

Figure 2:
Offset Correction



The purpose of the offset correction is, to equalize the positive and negative amplitudes of the signals. The following two steps accomplish this.

2.2.1 Offset Calculation

The offset is calculated out of the output signals minimum and maximum values according the following formula.

Equation 1: Offset Calculation

$$Offset = \frac{Signal_{Max} + Signal_{Min}}{2}$$

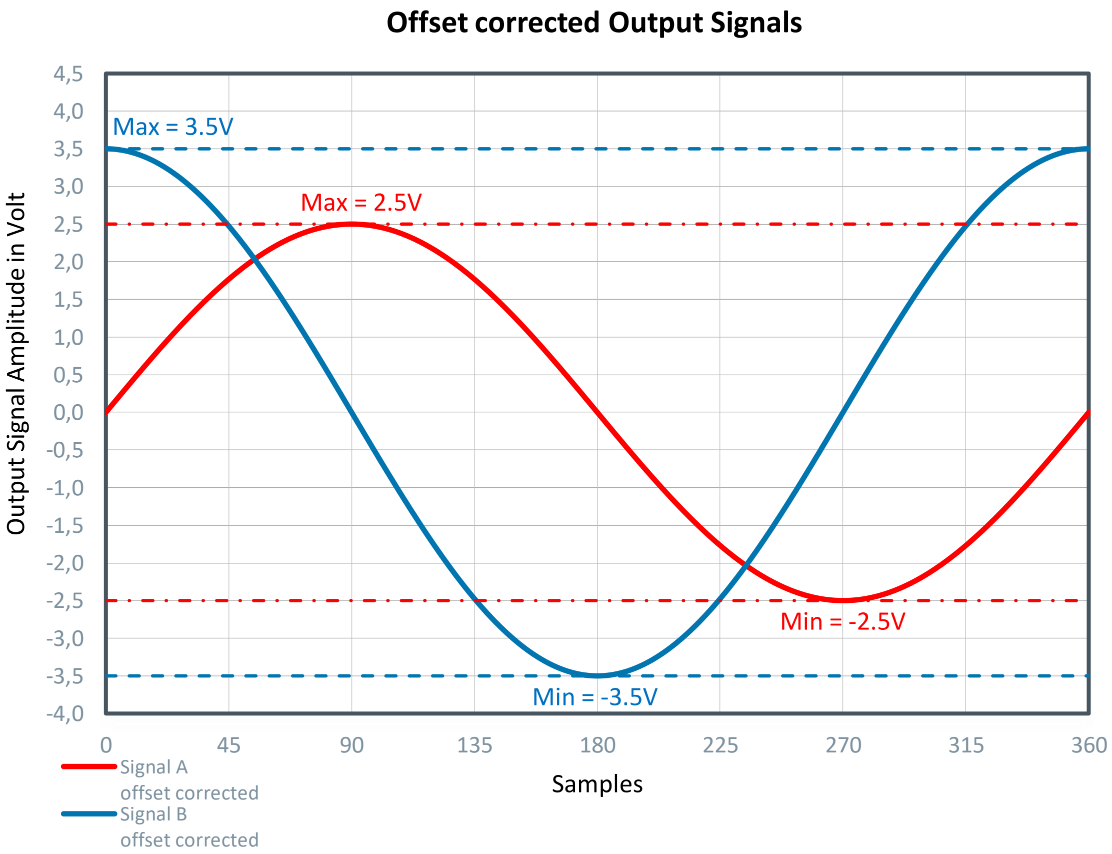
2.2.2 Offset corrected Output Signal

An offset corrected signal is obtained, by subtracting the previously calculated offset from the raw output signal.

Equation 2: Offset Correction

$$Signal_{Offset\_corrected} = Signal_{Raw} - Offset$$

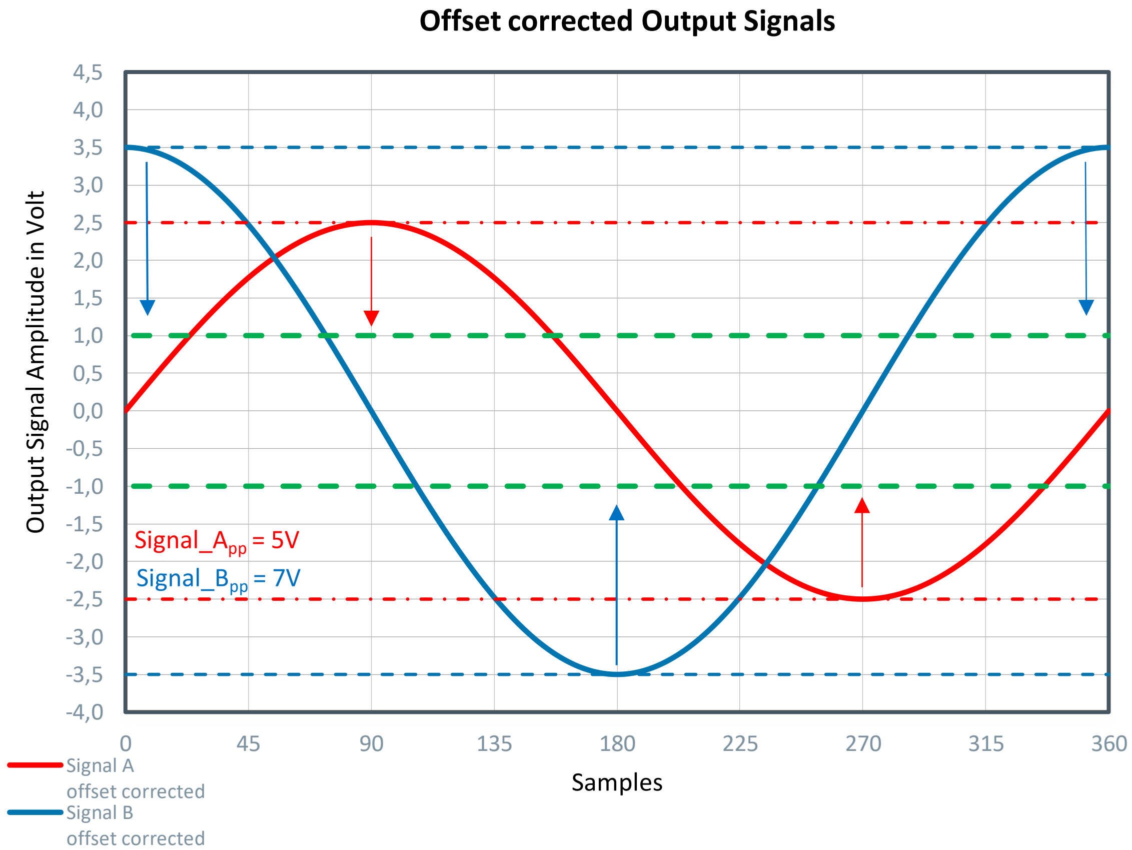
Figure 3:
Offset corrected Output Signals



As shown in Figure 3, the positive and negative peaks of each signal are equalized, but their individual amplitudes are still different.

2.3 Gain Correction

Figure 4:
Gain Correction



The purpose of the gain correction is, to equalize the amplitudes of both output signals. Furthermore, the amplitudes are normalized to 1V.

2.3.1 Gain corrected Output Signals

The following two formulas describe how the normalization of the output signals is carried out.

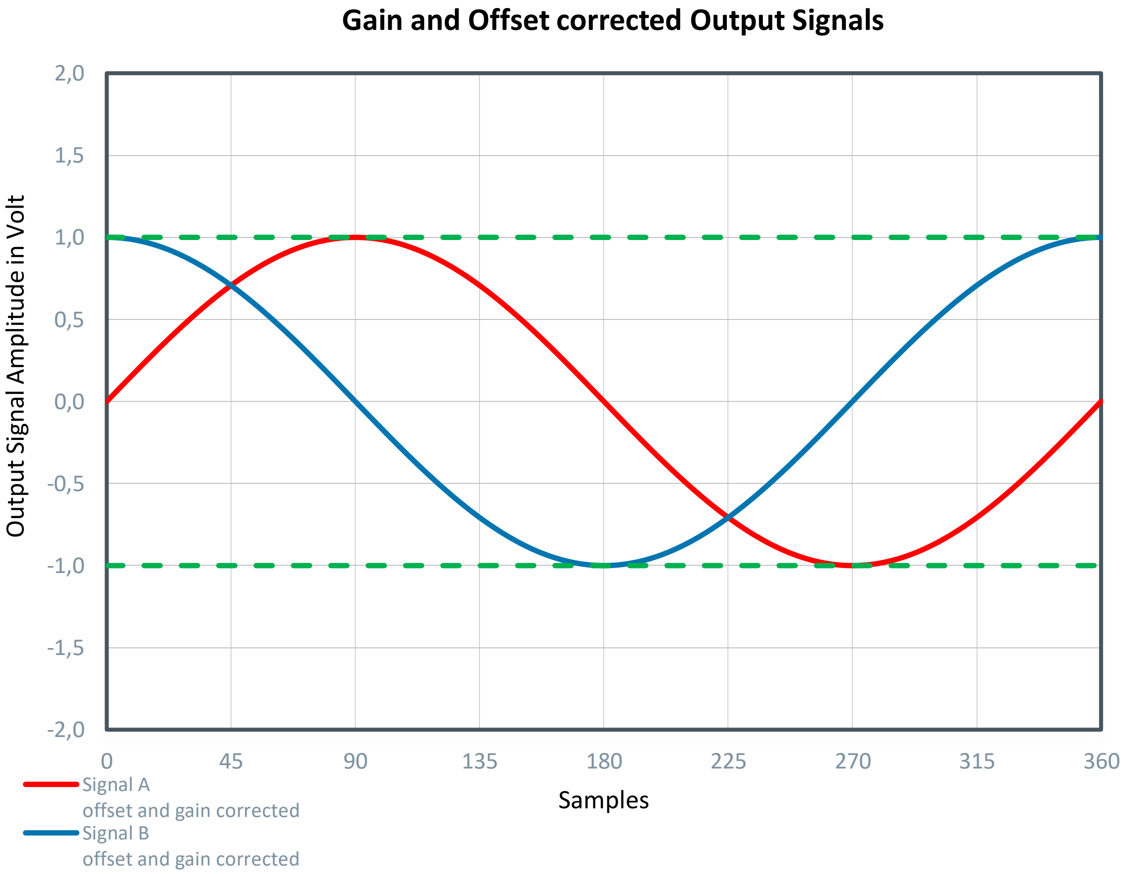
Equation 3: Gain Correction for Signal A

$$Signal\_A_{gain\_corrected} = \frac{Signal\_A}{\frac{Signal\_A_{pp}}{2}}$$

Equation 4: Gain Correction for Signal B

$$Signal\_B_{gain\_corrected} = \frac{Signal\_B}{\frac{Signal\_B_{pp}}{2}}$$

Figure 5: Gain and Offset corrected Output Signals



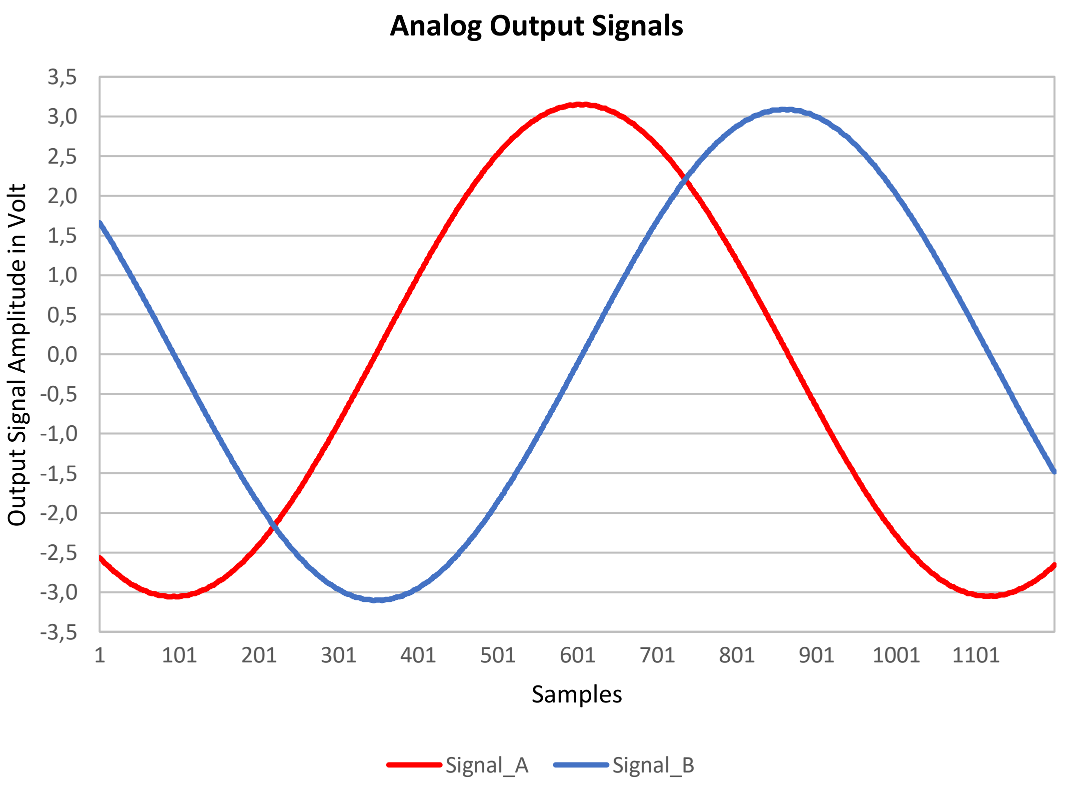
Now both signals do have the same amplitude and are free of any offset.

3 Angle Calculation

The following chapter explains how to calculate an angle out of two sinusoidal shaped signals.

3.1 Exemplary Output Signals

Figure 6:
Analog Output Signals



The diagram above shows two exemplary analog output signals.

3.2 Angle Calculation with atan2() function

To calculate an angle out of two sinusoidal shaped signals, the mathematical function atan2() is used. It is an angle function, which is part of most programming languages mathematic libraries. The atan2() function is also implemented in Microsoft Excel.

Equation 5: Angle Calculation with atan2()

$$ATAN2(Signal\_A; Signal\_B) = \varphi\ [rad]$$

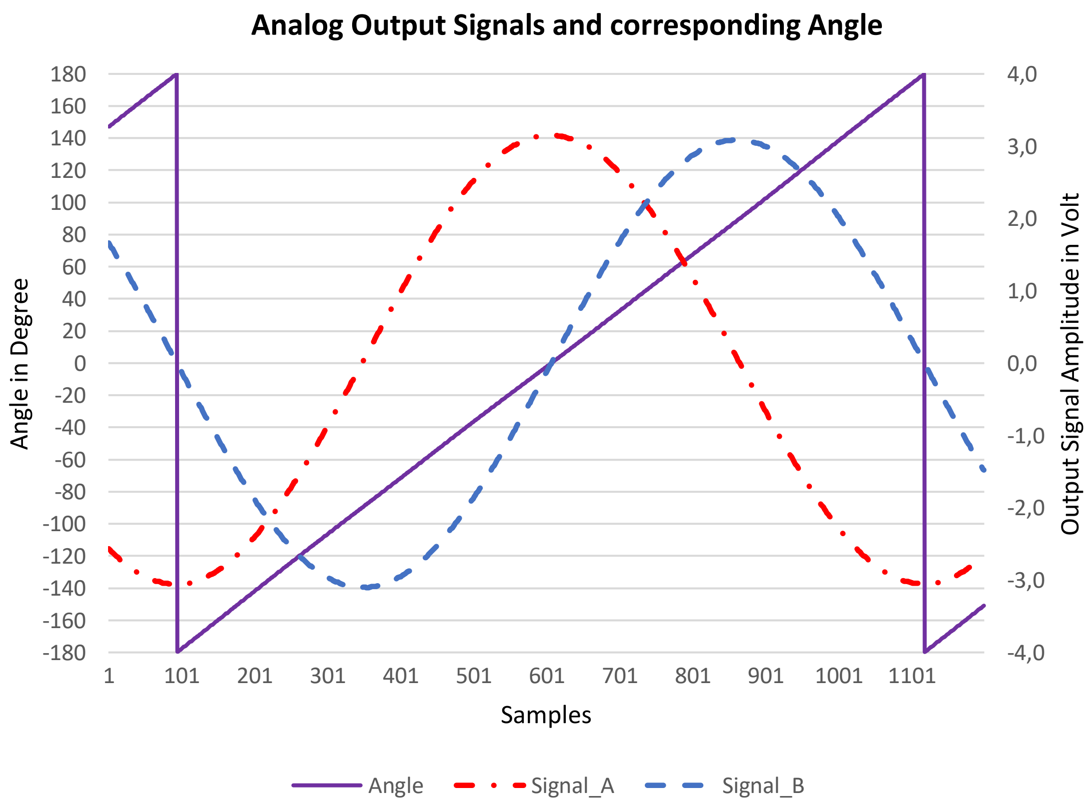
The result of the atan2() function is gives in radians. With the following formula the result is converted to degree.

Equation 6: Radians to Degree

$$\varphi\ [deg] = \varphi\ [rad] * (\frac{180}{\pi})$$

3.3 Resulting Angle

Figure 7:
Analog Output Signals and corresponding Angle



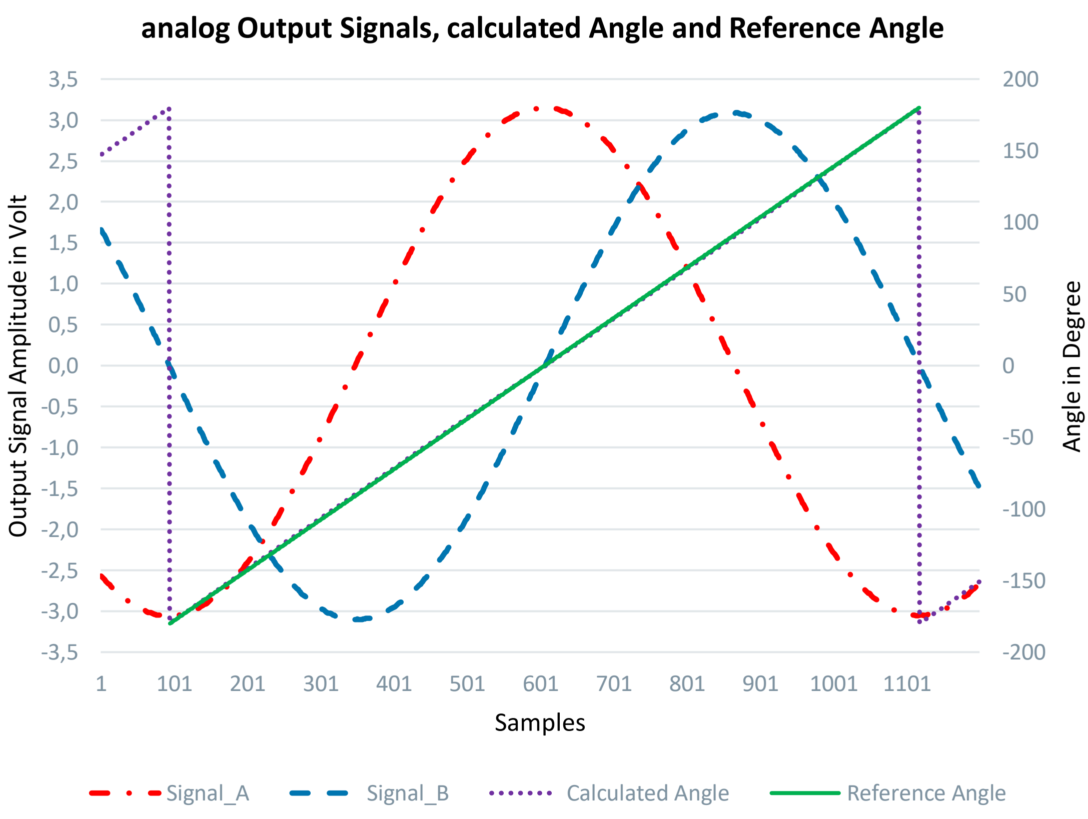
The resulting angle is illustrated in the diagram above. It ranges from -180° to +180°.

4 Integral Non-Linearity

The integral non-linearity, short INL, describes the deviation of the calculated angle from the best line of fit.

4.1 Reference Angle

Figure 8:
Analog Output Signals, Calculated Angle and Reference Angle



The best line of fit, here also called "*Reference Angle*" is illustrated in the diagram above.

4.2 INL Calculation

The first step when calculating the INL is, to build the difference between the calculated angle and the reference angle.

Equation 7: Difference between the actual and ideal angle

$$\Delta\varphi = \varphi_{Calc} - \varphi_{Ref}$$

with

φ_{Calc} … $Calculated\ Angle$

φ_{Ref} … $Reference\ Angle$

As a next step, the INL can be determined out of the maximum negative and the maximum positive deviation from the reference angle, according the following formula.

Equation 8: INL Calculation

$$INL = \frac{\Delta\varphi_{Max} - \Delta\varphi_{Min}}{2}$$

with

$\Delta\varphi_{Min}$ … $maximum\ negative\ deviation\ from\ reference\ angle$

$\Delta\varphi_{Max}$ … $maximum\ positive\ deviation\ from\ reference\ angle$

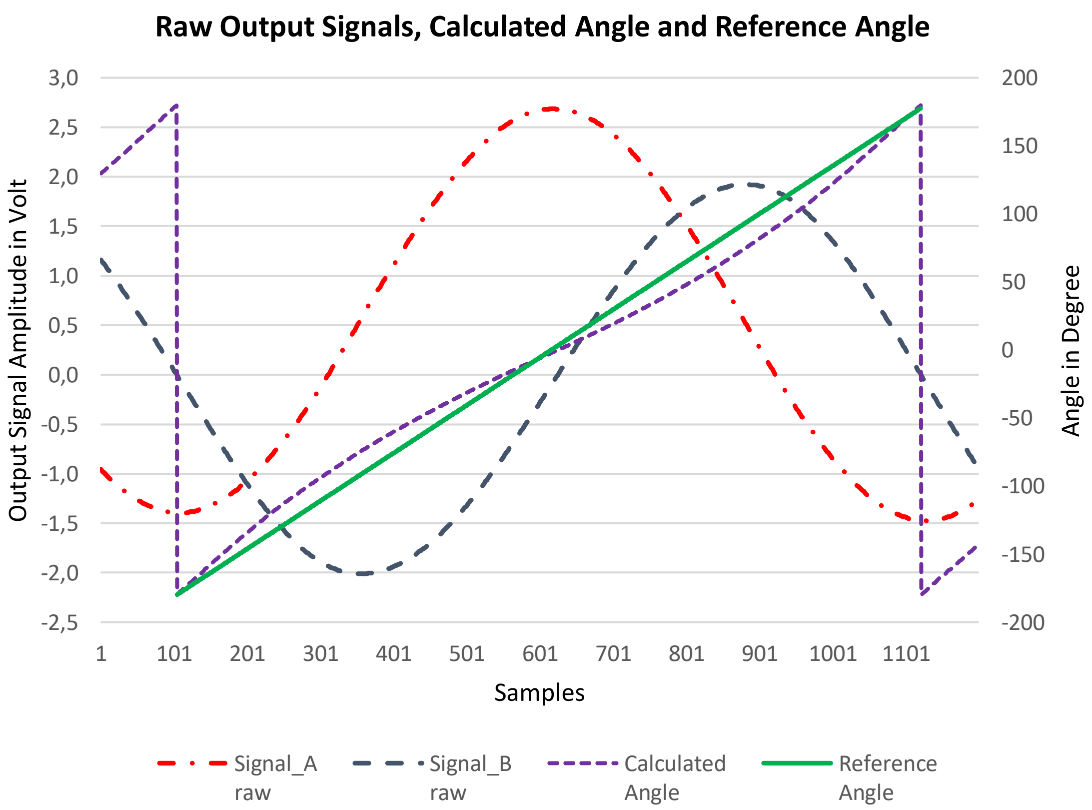
5 INL Comparison

The following Chapter clarifies how the INL improves, if it is calculated out of the gain and offset corrected signals rather than the raw signals. The calculations shown hereafter are based on actual measurement results. The complete raw data, as well as all calculations, can be found in the document *"Post_Processing_INL_Comparison.xlsx"*.

5.1 Raw Signals and corresponding Angle

Figure 9:

Output Signals before Gain and Offset Correction



As shown in Figure 9, the angle, which results from the raw output signals, is far off an ideal straight line. This will also have negative impact on the INL.

5.1.1 Calculation of INL_Raw

Equation 9: $\Delta\varphi_{Raw\_Min}$ and $\Delta\varphi_{Raw\_Max}$

$$\Delta\varphi_{Raw,Min} = -18.5201° \ldots \; maximum\ negative\ deviation\ from\ reference\ angle$$

$$\Delta\varphi_{Raw,Max} = +17.2392° \ldots \; maximum\ positive\ deviation\ from\ reference\ angle$$

Equation 10:INL_Raw

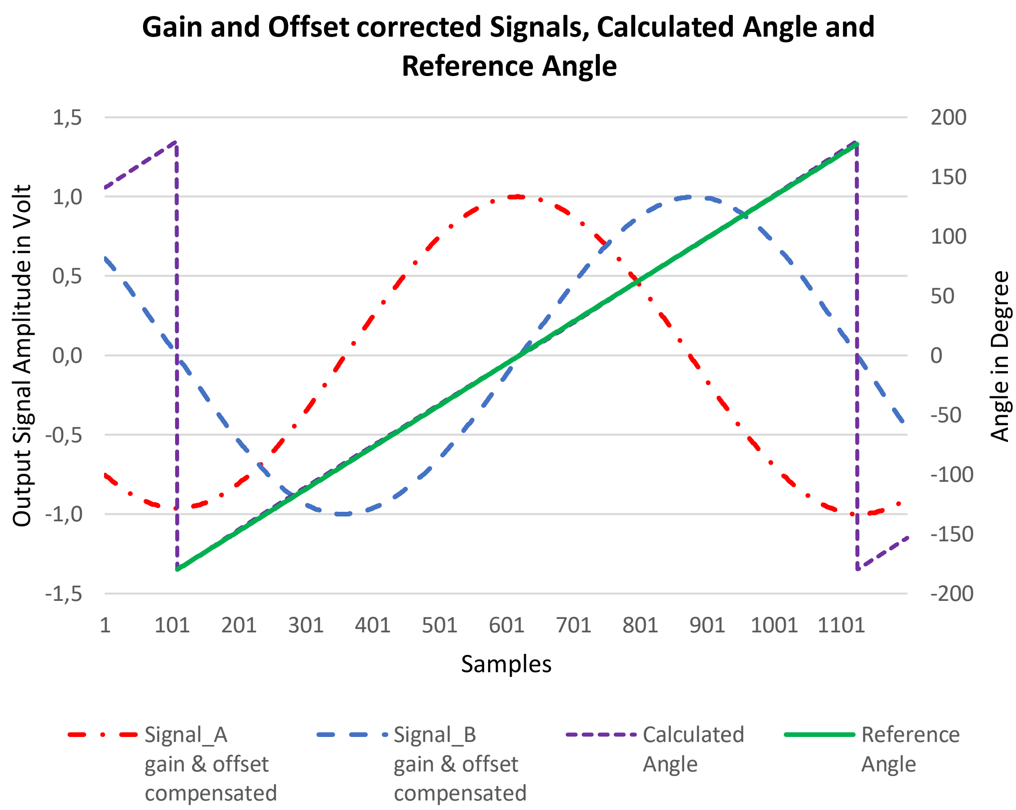
$$INL_{Raw} = \frac{\Delta\varphi_{Raw,Max} - \Delta\varphi_{Raw,Min}}{2}$$

$$INL_{Raw} = \frac{17.2392° - (-18.5201°)}{2} \triangleq \frac{35.7593°}{2}$$

$$INL_{Raw} = 17.8796°$$

5.2 Gain and Offset corrected Signals and corresponding Angle

Figure 10:
Output Signals after Gain and Offset Correction



As shown in figure 10, the angle, which results from the gain and offset corrected signals, nearly matches the reference angle.

5.2.1 Calculation of INL_Corrected

Equation 11: $\Delta\varphi_{Corr\_Min}$ and $\Delta\varphi_{Corr\_Max}$

$$\Delta\varphi_{Corr,Min} = -1.0896° \ldots\ maximum\ negative\ deviation\ from\ reference\ angle$$

$$\Delta\varphi_{Corr,Max} = +2.4989° \ldots\ maximum\ poitive\ deviation\ from\ reference\ angle$$

Equation 12: Calculation of INL_Corr

$$INL_{Corr} = \frac{\Delta\varphi_{Corr,Max} - \Delta\varphi_{Corr,Min}}{2}$$

$$INL_{Corr} = \frac{2.4989° - (-1.0896°)}{2} \triangleq \frac{3.5885°}{2}$$

$$INL_{Corr} = 1.7943°$$

6 Revision Information

Changes from previous version to current revision v0-01	Page
V0-01 Initial Version	all

- Page and figure numbers for the previous version may differ from page and figure numbers in the current revision.
- Correction of typographical errors is not explicitly mentioned.

7 Legal Information

RoHS Compliant & ams Green Statement

RoHS Compliant: The term RoHS compliant means that ams AG products fully comply with current RoHS directives. Our semiconductor products do not contain any chemicals for all 6 substance categories plus additional 4 substance categories (per amendment EU 2015/863), including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, RoHS compliant products are suitable for use in specified lead-free processes.

ams Green (RoHS compliant and no Sb/Br/Cl): ams Green defines that in addition to RoHS compliance, our products are free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material) and do not contain Chlorine (Cl not exceed 0.1% by weight in homogeneous material).

Important Information: The information provided in this statement represents ams AG knowledge and belief as of the date that it is provided. ams AG bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. ams AG has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. ams AG and ams AG suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

Headquarters

ams AG

Tobelbader Strasse 30

8141 Premstaetten

Austria, Europe

Tel: +43 (0) 3136 500 0

Please visit our website at www.ams.com

Buy our products or get free samples online at www.ams.com/Products

Technical Support is available at www.ams.com/Technical-Support

Provide feedback about this document at www.ams.com/Document-Feedback

For sales offices, distributors and representatives go to www.ams.com/Contact

For further information and requests, e-mail us at ams_sales@ams.com